

ELEC-E5522 Speech Processing Project

The goal of ELEC-E5522 is to familiarise the student with the design, implementation, evaluation and documentation of a selected speech processing topic. Together these parts constitute *a speech processing project*. As the project topic, the course addresses *speech coding based on linear prediction (LP)*.

The course is arranged as follows. First, an introduction lecture is given to recap the necessary background methodologies in LP that are needed in the project. Second, by working in teams of 2-4 members, the student conducts the project through the following steps:

- (1) The design and implementation of a speech codec (using MATLAB or Python) and given instructions.
- (2) The analysis and objective evaluation of the implemented codec.
- (3) The informal, small-scale subjective evaluation of the implemented speech coding method.
- (4) The documentation of the entire project.

The teams work in the project according to a pre-selected schedule consisting of four milestones. At each milestone, the team's progress is reported to the teacher in regular zoom / face-to-face **meetings** between the team members and the teacher.

Schedule:

- Introduction lecture (January 13, 2025, 14:15-16:00, in lecture room T5 - A133)
- Milestone 1: Sections 2-3.3 (completed by February 7, 2025)
- Milestone 2: Section 3.4 (completed by March 7, 2025)
- Milestone 3: Section 4 (completed by March 28, 2025)
- Milestone 4: Section 5 (completed by May 9, 2025)

The milestone meetings are an important part of the course, and they are targeted to generate discussions between students and the teacher on the course topic. If students need help to complete their work before a milestone meeting, they should contact the teacher to get instructions.

When a team has completed its work in a certain section as described above, the team should contact the teacher to book the corresponding milestone meeting date/time. NOTE: Students should only **use e-mail** in contacting the teacher (paavo.alku@aalto.fi).

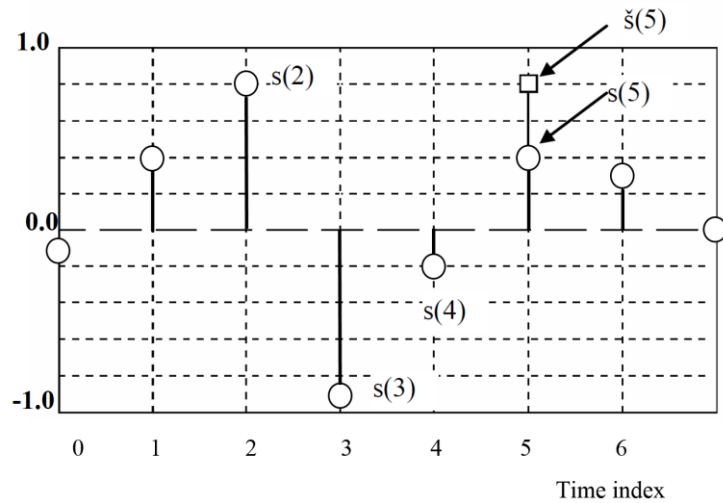
Pre-requisites: The course focus is on using **signal processing** methodologies in the selected topic of speech technology. Therefore, the student should have basic knowledge in speech processing (i.e. ELEC-E550 Speech Processing or similar course done) and in signal processing (i.e. ELEC-C5231 Introduction to Signal Processing or similar course done).

When teams have been organised by students, each team should mail **the names** of the team's members to the teacher.

ELEC-E5522 Speech Processing Project: The introduction lecture to linear prediction (LP)

1. Background

- Linear prediction (LP) (also called Linear Predictive Coding, LPC), is a parametric spectral estimation method that is among the most widely used tools in speech processing (Makhoul, 1975; Markel and Gray, 1976).
- Examples of speech technology areas where LP can be used:
 - speech coding (to compress speech)
 - speech and speaker recognition (in feature extraction)
 - speech synthesis (to express speech waveforms in a parametric form and to synthesise speech from these parameters)
 - speech analysis (in formant extraction, pitch estimation etc.)
- As a spectral estimation methods, LP belongs to autoregressive (AR) modelling
- Basic idea: Prediction of a speech sample as a linear combination of previous samples (see Fig. 1).



$$\hat{s}(5) = a(1)s(4) + a(2)s(3) + a(3)s(2)$$

Fig. 1. Speech sample at instant $n=5$, $s(5)$, is predicted from $p=3$ previous samples by multiplying them with coefficients $a(1)$, $a(2)$ and $a(3)$. Prediction denoted by $\hat{s}(5)$.

- Prediction is repeated for several consecutive samples (i.e. for a frame of samples) by using the same set of prediction coefficients $a(k)$, $1 \leq k \leq p$.
- The best set of coefficients is the one that yields the **minimum energy** of the prediction error.
- Energy-based optimization (i.e. mean squares error criterion) results in an optimization task that is mathematically easy to solve.
- Note: Prediction in LP takes advantage of three simple operations: addition, multiplication and delay, in other words, it uses an FIR (finite impulse response) filter structure.

2. Filter optimization

- By using p th order LP, the signal prediction can be formulated as:

$$\check{s}(n) = \sum_{k=1}^p a(k) s(n-k)$$

- Prediction error, the residual, can be expressed:

$$e(n) = s(n) - \check{s}(n) = s(n) - \sum_{k=1}^p a(k) s(n-k)$$

- The energy of the residual can be expressed as:

$$E = \sum_n e^2(n) = \sum_n \left[s(n) - \sum_{k=1}^p a(k)s(n-k) \right]^2$$

$$= \sum_n \left\{ s^2(n) - 2s(n) \sum_{k=1}^p a(k)s(n-k) + \left[\sum_{k=1}^p a(k)s(n-k) \right]^2 \right\}$$

- In order to minimize the residual energy, let us set the partial derivatives of E to zero:

$$\frac{\partial E}{\partial a(i)} = 0, \quad 1 \leq i \leq p$$

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$$\sum_n \left\{ -2s(n)s(n-i) + 2 \sum_{k=1}^p a(k)s(n-k)s(n-i) \right\} = 0, \quad 1 \leq i \leq p$$

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$$\sum_n s(n)s(n-i) = \sum_{k=1}^p a(k)s(n-k)s(n-i), \quad 1 \leq i \leq p$$

$$\sum_{k=1}^p a(k)\varphi(i,k) = \varphi(i,0), \quad \varphi(i,k) = \sum_n s(n-i)s(n-k) \quad 1 \leq i \leq p$$

- Equations above are known as the normal equations. They are a set of p equations with p unknowns (i.e. filter coefficients $a(k)$, $1 \leq k \leq p$).

- In equations above, the time span over which optimization is computed is left undefined.
- There are two LP variants which use different time spans for the optimization:
 - LP with the **autocorrelation criterion**: error minimization is computed in principle over $-\infty < n < \infty$
 - LP with the covariance criterion: minimization is computed over $0 \leq n \leq N-1$ where N denotes the frame size
- Autocorrelation criterion is by far the most widely used variant of LP because (1) it results in normal equations that can be solved with fast algorithms and (2) the resulting optimal FIR (LP inverse filter) is guaranteed to be **minimum phase** (i.e. its roots are always inside the unit circle in the z-domain which implies that the inverse of the FIR is a stable all-pole filter).
- In the following, we will only discuss LP with the autocorrelation criterion.

- Infinite time span cannot be treated in real-life applications. Therefore, infinite time duration is changed to finite by assuming that signal is zero outside the frame (i.e. speech waveform is **windowed** into frames of finite length and the waveform is assumed to be zero outside the frame).

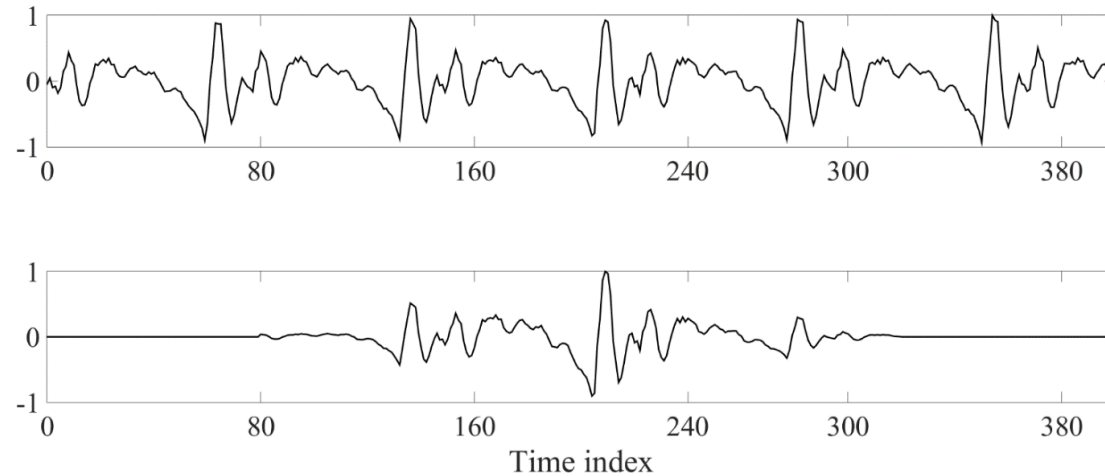


Fig. 2. Signal frame (bottom) windowed from continuous speech (top). Hamming window, frame position $80 \leq n \leq 320-1$.

- If frame length is N samples, the residual of a p th order LP is non-zero over $0 \leq n \leq N-1+p$. Therefore, we can write the normal equations as follows in matrix notation:

$$\varphi(i, k) = R(i - k) = R(k - i) = \sum_{j=0}^{N-1-(i-k)} s(j)s(j + i - k)$$

$$\mathbf{R} \cdot \mathbf{A} = \mathbf{R}'$$

$\Rightarrow$

$$\mathbf{A} = \mathbf{R}^{-1} \cdot \mathbf{R}'$$

$$\mathbf{R} = \begin{bmatrix} R(0) & R(1) & \dots & R(p-1) \\ R(1) & R(0) & \dots & R(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ R(p-1) & R(p-2) & \dots & R(0) \end{bmatrix}$$

$$\mathbf{A} = (a(1), a(2), \dots, a(p))^T$$

$$\mathbf{R}' = (R(1), R(2), \dots, R(p))^T$$

Summary of the main parts of LP analysis based on the autocorrelation criterion:

1. Select the order of prediction (p). As rule of thumb, p should be selected as the sampling frequency in kHz added by a small integer (Markel and Gray, 1976). For example, if signal is sampled with 8 kHz, a proper prediction order is $p=10$ or $p=12$.
2. Select a value for the frame length (N) and the window type. A good choice for N is a value that guarantees that there are a few fundamental cycles of voiced speech within the frame, yet not too many in order to keep the signal stationary. Hence, a typical window length is 20-30 ms which corresponds to 160-240 samples when $f_s=8$ kHz. Windowing is recommended to be done with windows such as Hamming or Hann (i.e. not rectangular).
3. Window the signal to get N windowed samples into the frame.
4. Compute autocorrelation terms $R(0)$ - $R(p)$.
5. Solve matrix **A**. Due to the fact that the autocorrelation matrix in **R** is symmetric (= columns correspond to rows) and Toeplitz (= same element in diagonals), matrix inversion can be computed with the Levison-Durbin recursion which is computationally efficient.
6. Form a digital FIR filter from the solved coefficients:

$$A(z) = 1 - \sum_{k=1}^p a(k)z^{-k}$$

7. Filter the original speech frame with $A(z)$ to obtain the residual.

3. LP filtering

- LP can be considered to consist of two parts:
 - (1) LP analysis: separation of a speech frame into a filter (FIR, LP-inverse filter) and time-domain residual
 - (2) LP synthesis: synthesis of the speech signal from the residual and the LP-filter (IIR, all-pole)
- Since residual is noise-like and spectrally flat, LP-filter forms an all-pole match to the speech spectrum

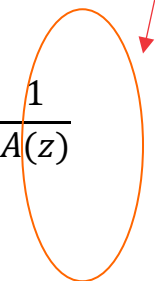
LP analysis:

Time domain:
$$e(n) = s(n) - \sum_{k=1}^p a(k)s(n-k)$$

z-domain:
$$E(z) = S(z) \cdot A(z)$$

LP synthesis:

$$s(n) = e(n) + \sum_{k=1}^p a(k)s(n-k)$$

$$S(z) = E(z) \cdot \frac{1}{A(z)}$$


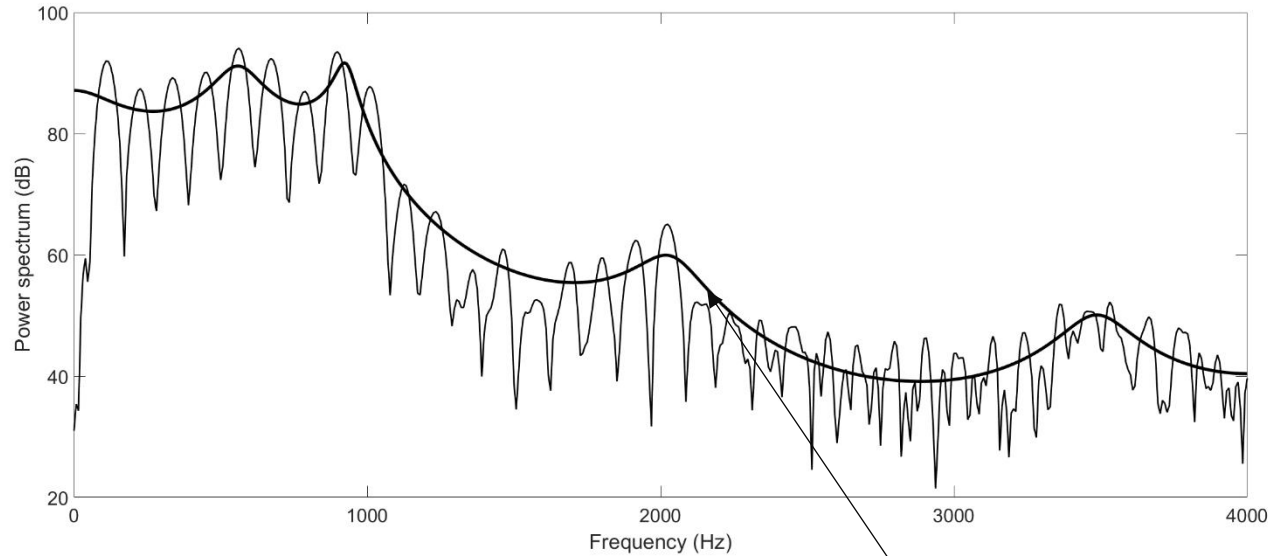


Fig. 3. Spectra computed from a vowel sound ([a]):
FFT-spectrum (thin) and LP-spectrum (thick) ($p=10$).

$$H(z) = \frac{\sigma}{A(z)} = \frac{\sigma}{1 - \sum_{k=1}^p a(k)z^{-k}}$$

4. LP information

4.1 General

- Most speech technology applications of LP take advantage of the LP filter (e.g. compression, feature extraction, formant trajectory modelling etc.).
- LP filter (i.e. p th-order all-pole type of IIR filter) contains the essence of LP analysis and the data needed to define this filter are called LP information.
- In filter optimization, LP information is expressed in the form of p filter coefficients ($a(k)$, $1 \leq k \leq p$) of a direct form filter structure.
- In many applications, LP information needs to be processed (i.e. quantized, predicted using, e.g., DNNs etc.) which changes the computed coefficients.

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LP synthesis filter, which is guaranteed to be stable in its **original form** when computed with the autocorrelation criterion, might become **unstable**.

- In addition to using the direct form filter taps, there are alternative ways to express the LP information:
 - using a lattice filter structure with filter coefficients called **reflection coefficients**
 - using the line spectrum pair (LSP) decomposition (Soong and Juang, 1984)

In this course, the former way will be used.

4.2 Reflection coefficients

- Solving the normal equations (section 2) with the Levinson-Durbin recursion can be expressed:

$$E^{(0)} = R(0)$$

$$k_i = \left\{ R(i) - \sum_{j=1}^{i-1} \alpha_j^{(i-1)} R(|i-j|) \right\} / E^{(i-1)}$$

$$\alpha_i^{(i)} = k_i$$

$$\alpha_j^{(i)} = \alpha_j^{(i-1)} - k_i \alpha_{i-j}^{(i-1)} \quad 1 \leq j \leq i-1$$

$$E^{(i)} = (1 - k_i^2) E^{(i-1)}$$

- The set of equations above (except the first line) are repeated recursively with $i = 1, 2, \dots, p$ resulting in LP coefficients:

$$a(i) = \alpha_i^{(p)}, 1 \leq i \leq p$$

- As by-product, the recursion yields reflection coefficients

$$k_i, 1 \leq i \leq p$$

- Reflection coefficients enable presenting the LP inverse filter in the lattice filter form (Oppenheim and Schaffer, 1991). Fig. 2 shows two 3rd order LP inverse filters, one in direct form structure (upper) and the other one (lower) in lattice structure.
- Using reflection coefficients is motivated by the following fact:

$$A(z) \text{ is minimum phase} \quad A(z) = 1 - \sum_{k=1}^p a(k)z^{-k}$$

$\Leftrightarrow$

all p reflection coefficients are between -1.0 and 1.0

- The property above helps in checking stability of $1/A(z)$ when LP information is quantized.
- Reflection coefficients can be quantized using, for example, simple uniform quantization (Fig. 5).

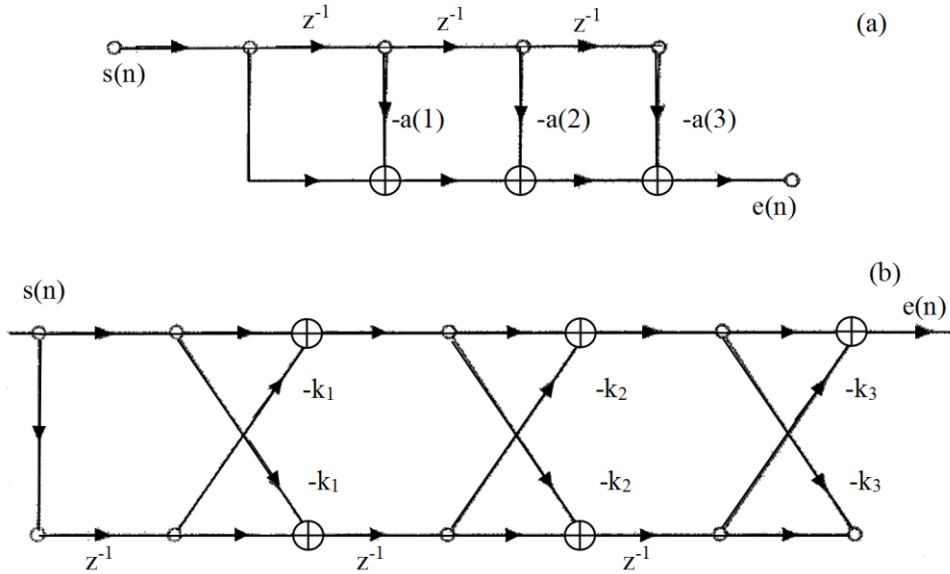


Fig. 4. Third order ($p=3$) LP inverse filter in (a) direct form structure and in (b) lattice structure.

Conversion between direct form filter coefficients and reflection coefficients

Conversions between direct form and lattice filters are given in Oppenheim and Schaffer (1991). By referring to their textbook, let us change our earlier direct form filter tap notation from $a(k)$ to a_m . The transfer function $A(z)$ can then be denoted as $A(z) = 1 - \sum_{m=1}^p a_m z^{-m}$. The filter taps of $A(z)$ are denoted in the following as $a_m = a_m^{(p)}, m = 1, 2, \dots, p$

1) Conversion from direct form taps to reflection coefficients:

1.1) $k_p = a_p^{(p)}$ (i.e. the last reflection coefficient equals the last direct form tap)

1.2) Repeat the following for $i=p, p-1, \dots, 1$

$$k_i = a_i^{(i)}$$

$$a_m^{(i-1)} = \frac{a_m^{(i)} + k_i a_{i-m}^{(i)}}{1 - k_i^2}, m = 1, 2, \dots, (i-1)$$

The reflection coefficients are obtained in the recursion above as $k_i, i = 1, 2, \dots, p$

2) Conversion from reflection coefficients to direct form taps

Repeat the following for $i=1, 2, \dots, p$

$$a_i^{(i)} = k_i$$

$$a_m^{(i)} = a_m^{(i-1)} - k_i a_{i-m}^{(i-1)}, m = 1, 2, \dots, (i-1)$$

The direct form taps are given by $a_m = a_m^{(p)}, m = 1, 2, \dots, p$

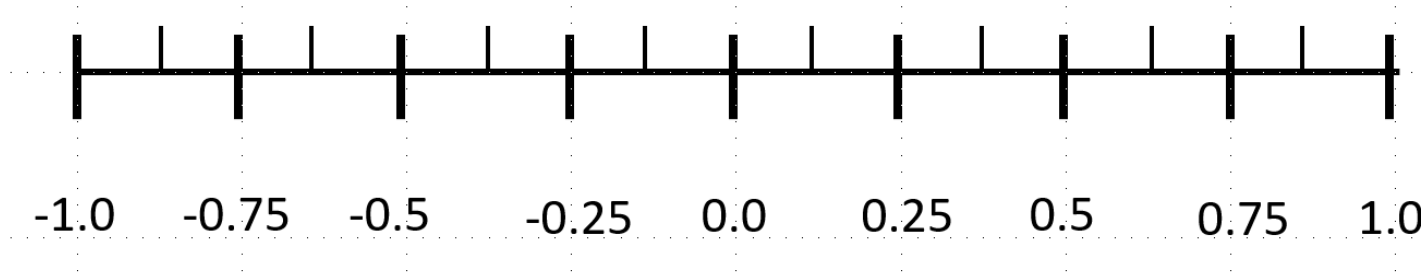


Fig. 5. An uniform quantization scale to quantize reflection coefficients with 3 bits. The scale ranges between the smallest (-1.0) and largest (1.0) possible value. Eight intervals of equal length are separated by the thick lines. In each interval, all reflection coefficient values are quantized to the middle value shown by the thin lines.

References:

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Makhoul, J. (1975) Linear Prediction: A tutorial review, Proc. IEEE, Vol. 63, pp. 561-580.

Markel, J.D., Gray, A.H., Jr. (1976). Linear Prediction of Speech, Springer-Verlag, 1976.

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